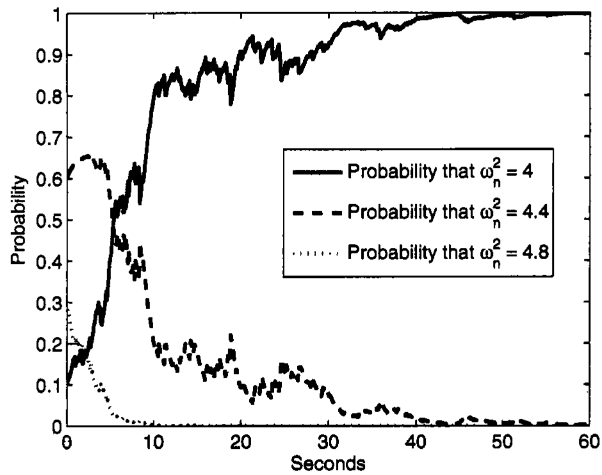

Figure 10.2 Parameter probabilities for the multiple-model Kalman filter for Example 10.1. The true parameter value is 4, and the filter converges to the correct parameter after a few seconds.

10.3 REDUCED-ORDER KALMAN FILTERING

If a user wants to estimate only a subset of the state vector, then a reduced-order filter can be designed. This can be the case, for example, in a real-time application where computational effort is a main consideration. Even in off-line applications, some types of problems (e.g., weather forecasting) can involves tens of thousands of states, which naturally motivates reduced-order filtering as a means to reduce computational effort [Pha98, Bal01].

Various approaches to reduced-order filtering have been proposed over the years. For example, if the dynamic model of the underlying system can be reduced to a lower-order model that approximates the full-order model, then the reduced-order model can form the basis of a normally designed Kalman filter [Kel99]. This is the approach taken in [Gli94, Sot99] for motor state estimation, in [Sim69, Ara94] for navigation system alignment, in [Bur93, Pat98] for image processing, and in [Cha96] for audio processing. If some of the states are not observable then the Kalman filter Riccati equation reduces to a lower-order equation [Yon80]. Reduced-order filtering can be implemented by approximating the covariance with a lower-rank SVD-like decomposition [Pha98, Bal01]. If some of the measurements are noise free, or if there are known equality constraints between some of the states, then the Kalman filter is a filter with an order that is lower than the underlying system [Bry65, Hae98] as discussed in Section 7.5.1 of this book. Optimal reduced-order filters are obtained from first principles in [Ber85, Nag87]. A more heuristic approach to reduced-order filtering is to decouple portions of the matrix multiplications in the Kalman filter equations [Chu87]. In this section we will present two different approaches to reduced-order filtering.

10.3.1 Anderson's approach to reduced-order filtering

Anderson and Moore [And79] suggest a framework for reduced-order filtering that is fully developed in [Sim06a] and in this section. This approach is based on the idea that we do not always need to estimate all of the states of a system. Sometimes, with a system that has n states, we are interested only in estimating m linear combinations of the states, where $m < n$. In this case, it stands to reason that we could devise a filter with an order less than n that estimates the m linear combinations that we are interested in. Suppose our state space system is given as

$$\begin{aligned} \bar{x}_{k+1} &= \bar{F}\bar{x}_k + \bar{G}w_k \\ y_k &= \bar{H}\bar{x}_k + v_k \end{aligned} \tag{10.38}$$

We desire to estimate the following m linear combinations of the state: $T_1^T\bar{x}$, $T_2^T\bar{x}$, $\cdots$, $T_m^T\bar{x}$, where each T_i^T is a row vector. Define the $n \times n$ matrix T as

$$T = \begin{bmatrix} T_1^T \\ \vdots \\ T_m^T \\ S \end{bmatrix} \tag{10.39}$$

where S is arbitrary as long as it makes T a nonsingular $n \times n$ matrix. Now perform the state transformation

$$x = T\bar{x} \tag{10.40}$$

This means that $\bar{x} = T^{-1}x$. From these relationships we can obtain a state space description of the system in terms of the new state as follows:

$$\begin{aligned} T^{-1}x_{k+1} &= \bar{F}T^{-1}x_k + \bar{G}w_k \\ x_{k+1} &= T\bar{F}T^{-1}x_k + T\bar{G}w_k \\ &= Fx_k + Gw_k \\ y_k &= \bar{H}T^{-1}x_k + v_k \\ &= Hx_k + v_k \end{aligned} \tag{10.41}$$

where F, G, and H are defined by the above equations. Remember that our goal is to estimate the first m elements of x, which we will denote as $\tilde{x}$. We therefore partition x as follows:

$$x = \begin{bmatrix} \tilde{x} \\ \tilde{\tilde{x}} \end{bmatrix} \tag{10.42}$$

We can then write equations for $\tilde{x}_{k+1}$, $\tilde{\tilde{x}}_{k+1}$, and y_k as follows:

$$\begin{aligned} \tilde{x}_{k+1} &= F_{11}\tilde{x}_k + F_{12}\tilde{\tilde{x}}_k + G_1 w_k \\ \tilde{\tilde{x}}_{k+1} &= F_{21}\tilde{x}_k + F_{22}\tilde{\tilde{x}}_k + G_2 w_k \\ y_k &= H_1\tilde{x}_k + H_2\tilde{\tilde{x}}_k + v_k \end{aligned} \tag{10.43}$$

where the F_{ij}, G_i, and H_i matrices are appropriately dimensioned partitions of F, G, and H. Now we propose the following form for the one-step *a posteriori* estimator of $\tilde{x}$:

$$\hat{\tilde{x}}_{k+1}^+ = F_{11}\hat{\tilde{x}}_k^+ + K_k(y_{k+1} - H_1F_{11}\hat{\tilde{x}}_k^+) \tag{10.44}$$

This predictor/corrector form for the estimate of $\tilde{x}$ is very similar to the predictor/corrector form of the standard Kalman filter. The estimation error is given as follows:

$$\begin{aligned}
e_{k+1} &= \tilde{x}_{k+1} - \hat{\tilde{x}}_{k+1}^{+} \\
&= F_{11}(\tilde{x}_k - \hat{\tilde{x}}_k^{+}) + F_{12}\tilde{\tilde{x}}_k + G_1 w_k - K_k(y_{k+1} - H_1 F_{11}\hat{\tilde{x}}_k^{+}) \\
&= (I - K_k H_1)F_{11}e_k + [F_{12} - K_k(H_1F_{12} - H_2F_{22})]\tilde{\tilde{x}}_k - \\
&\quad K_kH_2F_{21}\tilde{x}_k - K_k v_{k+1} + [G_1 - K_k(H_1G_1 + H_2G_2)]w_k
\end{aligned} \tag{10.45}$$

Now we will introduce the following notation for various covariance matrices:

$$\begin{aligned}
P_k &= E(e_k e_k^T) \\
\tilde{P}_k &= E(\tilde{x}_k \tilde{x}_k^T) \\
\tilde{\tilde{P}}_k &= E(\tilde{\tilde{x}}_k \tilde{\tilde{x}}_k^T) \\
\Sigma_k &= E(\tilde{x}_k \tilde{\tilde{x}}_k^T) \\
\tilde{\Pi}_k &= E(\hat{\tilde{x}}_k \tilde{x}_k^T) \\
\tilde{\tilde{\Pi}}_k &= E(\hat{\tilde{x}}_k \tilde{\tilde{x}}_k^T)
\end{aligned} \tag{10.46}$$

With this notation and the equations given earlier in this section, we can obtain the following expressions for these covariances:

$$\begin{aligned}
\tilde{P}_{k+1} &= F_{11}\tilde{P}_kF_{11}^T + (F_{11}\Sigma_kF_{12}^T) + (\cdots)^T + F_{12}\tilde{\tilde{P}}_kF_{12}^T + G_1Q_kG_1^T \\
\tilde{\tilde{P}}_{k+1} &= F_{21}\tilde{P}_kF_{21}^T + (F_{21}\Sigma_kF_{22}^T) + (\cdots)^T + F_{22}\tilde{\tilde{P}}_kF_{22}^T + G_2Q_kG_2^T \\
\Sigma_{k+1} &= F_{11}\tilde{P}_kF_{21}^T + F_{11}\Sigma_kF_{22}^T + F_{12}\tilde{\tilde{P}}_kF_{22}^T + G_1Q_kG_2^T \\
\tilde{\Pi}_{k+1} &= (I - K_kH_1)F_{11}(\tilde{\Pi}_kF_{11}^T + \tilde{\tilde{\Pi}}_kF_{12}^T) + \\
&\quad K_k(H_1F_{11} + H_2F_{21})(\tilde{P}_kF_{11}^T + \Sigma_kF_{12}^T) + \\
&\quad K_k(H_1F_{12} + H_2F_{22})(\Sigma_k^TF_{11}^T + \tilde{\tilde{P}}_kF_{12}^T) + \\
&\quad K_k(H_1G_1 + H_2G_2)Q_kG_1^T \\
\tilde{\tilde{\Pi}}_{k+1} &= (I - K_kH_1)F_{11}(\tilde{\Pi}_kF_{21}^T + \tilde{\tilde{\Pi}}_kF_{22}^T) + \\
&\quad K_k(H_1F_{11} + H_2F_{21})(\tilde{P}_kF_{21}^T + \Sigma_kF_{22}^T) + \\
&\quad K_k(H_1F_{12} + H_2F_{22})(\Sigma_k^TF_{21}^T + \tilde{\tilde{P}}_kF_{22}^T) + \\
&\quad K_k(H_1G_1 + H_2G_2)Q_kG_2^T \\
P_{k+1} &= (I - K_kH_1)F_{11}P_kF_{11}^T(I - K_kH_1)^T + \\
&\quad \Big[(I - K_kH_1)F_{11}(\Sigma_k - \tilde{\tilde{\Pi}}_k)Y_{k+1}\Big] + \Big[\cdots\Big]^T + \\
&\quad \Big[(I - K_kH_1)F_{11}(\tilde{\Pi}_k - \tilde{P}_k)F_{21}^TH_2^TK_k^T\Big] + \Big[\cdots\Big]^T + \\
&\quad Y_{k+1}^T\tilde{\tilde{P}}_kY_{k+1} - \big(Y_{k+1}^T\Sigma_k^TF_{21}^TH_2^TK_k^T\big) + \big(\cdots\big)^T + \\
&\quad K_kH_2F_{21}\tilde{P}_kF_{21}^TH_2^TK_k^T + K_kR_{k+1}K_k^T + \\
&\quad [G_1 - K_k(H_1G_1 + H_2G_2)]Q_k[G_1 - K_k(H_1G_1 + H_2G_2)]^T
\end{aligned} \tag{10.47}$$

where Y_k is defined as

$$Y_k = [F_{12} - K_k(H_1F_{12} + H_2F_{22})]^T \tag{10.48}$$

Now we can find the optimal reduced-order gain K_k at each time step as follows:

$$\begin{aligned} K_k &= \text{argmin } \text{Tr} P_{k+1} \\ \frac{\partial \text{Tr} P_{k+1}}{\partial K_k} &= 0 \end{aligned} \tag{10.49}$$

In order to compute the partial derivative we have to remember from Section 1.1.3 that

$$\begin{aligned} \frac{\partial \text{Tr}(ABA^T)}{\partial A} &= AB + AB^T \\ \frac{\partial \text{Tr}(AB)}{\partial A} &= B^T \\ \frac{\partial \text{Tr}(BA^T)}{\partial A} &= B \end{aligned} \tag{10.50}$$

Armed with these tools we can compute the partial derivative of Equation 10.49 and set it equal to zero to obtain

$$K_k = A_k^{-1} B_k \tag{10.51}$$

where A_k and B_k are given as follows:

$$\begin{aligned} A_k &= H_1 F_{11} P_k F_{11}^T H_1^T + \left[H_1 F_{11} (\Sigma_k - \tilde{\tilde{\Pi}}_k)(H_1 F_{12} + H_2 F_{22})^T \right] + \\ & \quad \left[\cdots \right]^T + \left[H_{11} F_{11} (\tilde{P}_k - \tilde{\Pi}_k) F_{21}^T H_2^T \right] + \left[\cdots \right]^T + \\ & \quad (H_1 F_{12} + H_2 F_{22}) \tilde{\tilde{P}}_k (H_1 F_{12} + H_2 F_{22})^T + \\ & \quad \left[(H_1 F_{12} + H_2 F_{22}) \Sigma_k^T F_{21}^T H_2^T \right] + \left[\cdots \right]^T + H_2 F_{21} \tilde{P}_k F_{21}^T H_2^T + \\ & \quad R_{k+1} + (H_1 G_1 + H_2 G_2) Q_k (H_2 G_1 + H_2 G_2)^T \\ B_k &= \left(F_{11} P_k + F_{12} \Sigma_k^T - F_{12} \tilde{\tilde{\Pi}}^T \right) F_{11}^T H_1^T + \\ & \quad \left(F_{11} \Sigma_k - F_{11} \tilde{\tilde{\Pi}}_k + F_{12} \tilde{\tilde{P}}_k \right) (H_1 F_{12} + H_2 F_{22})^T + \\ & \left(F_{11} \tilde{P}_k - F_{11} \tilde{\Pi}_k + F_{12} \Sigma_k^T \right) F_{21}^T H_2^T + G_1 Q_k (H_1 G_1 + H_2 G_2)^T \end{aligned} \tag{10.52}$$

Equation (10.51) ends up being a long and complicated expression for the reduced-order gain. In fact, this reduced-order filter is probably more computationally expensive than the full-order filter (depending on the values of m and n). However, if the gain of the reduced-order filter converges to steady state, then it can be computed off-line to obtain savings in real-time computational cost and memory usage. However, note that the reduced-order filter may not be stable, even if the full-order Kalman filter is stable.

■ EXAMPLE 10.2

Suppose we are given the following system:

$$x_{k+1} = \begin{bmatrix} 0.9 & 0.1 \\ 0.2 & 0.7 \end{bmatrix} x_k + \begin{bmatrix} 1 \\ 0 \end{bmatrix} w_k$$

$$\begin{aligned} y_k &= \begin{bmatrix} 0 & 1 \end{bmatrix} x_k + v_k \\ w_k &\sim (0, 0.1) \\ v_k &\sim (0, 1) \end{aligned} \tag{10.53}$$

We want to find a reduced-order estimator of the first element of x. In this example the reduced-order gain of Equation (10.51) converges to a steady-state value after about 80 time steps. The estimation-error variance of the reduced-order filter converges to a value that is about 10% higher than the estimation-error variance of the full-order filter for the first state, as shown in Figure 10.3. The estimation error for the reduced-order filter and the full-order filter is shown in Figure 10.3 for a typical simulation. In this example, the standard deviation of the estimation error was 0.46 for the full-order filter and 0.50 for the reduced-order filter. The steady-state full-order estimator is given as follows:

$$\begin{aligned} \hat{x}_{k+1}^- &= \begin{bmatrix} 0.9 & 0.1 \\ 0.2 & 0.7 \end{bmatrix} \hat{x}_k^+ \\ \hat{x}_k^+ &= \hat{x}_k^- + K \left(y_k - \begin{bmatrix} 0 & 1 \end{bmatrix} \hat{x}_k^- \right) \\ K &= \begin{bmatrix} 0.1983 \\ 0.1168 \end{bmatrix} \end{aligned} \tag{10.54}$$

The steady-state reduced-order estimator is given as follows:

$$\begin{aligned} \hat{\tilde{x}}_{k+1}^+ &= 0.9\hat{\tilde{x}}_k^+ + K_r \left[y_{k+1} - (0)(0.9)\hat{\tilde{x}}_k^+ \right] \\ &= 0.9\hat{\tilde{x}}_k^+ + K_r y_{k+1} \\ K_r &= 0.1420 \end{aligned} \tag{10.55}$$

$\nabla\nabla\nabla$

10.3.2 The reduced-order Schmidt–Kalman filter

Stanley Schmidt's approach to reduced-order filtering can be used if the states are decoupled from each other in the dynamic equation [Sch66, Bro96, Gre01]. This happens, for instance, if colored measurement noise is accounted for by augmenting the state vector (see Section 7.2.2). In fact, satellite navigation with colored measurement noise was the original motivation for this approach.

Suppose we have a system in the form

$$\begin{aligned} \begin{bmatrix} \tilde{x}_{k+1} \\ \tilde{\tilde{x}}_{k+1} \end{bmatrix} &= \begin{bmatrix} F_1 & 0 \\ 0 & F_2 \end{bmatrix} \begin{bmatrix} \tilde{x}_k \\ \tilde{\tilde{x}}_k \end{bmatrix} + \begin{bmatrix} \tilde{w}_k \\ \tilde{\tilde{w}}_k \end{bmatrix} \\ \tilde{w}_k &\sim (0, Q_1) \\ \tilde{\tilde{w}}_k &\sim (0, Q_2) \\ y_k &= \begin{bmatrix} H_1 & H_2 \end{bmatrix} \begin{bmatrix} \tilde{x}_k \\ \tilde{\tilde{x}}_k \end{bmatrix} + v_k \\ v_k &\sim (0, R) \end{aligned} \tag{10.56}$$

We want to estimate $\tilde{x}_k$ but we do not care about estimating $\tilde{\tilde{x}}_k$. Suppose we use a Kalman filter to estimate the entire state vector. The estimation-error covariance

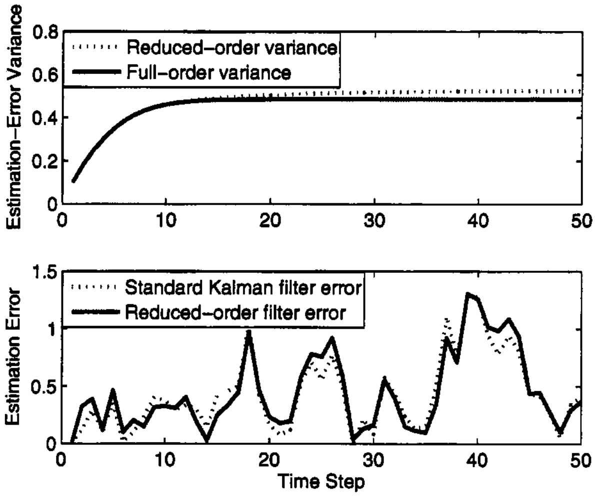

Figure 10.3 Results for Example 10.2. The top figure shows the analytical estimation-error variances for the first state for the full-order filter and the reduced-order filter. As expected, the reduced-order filter has a higher estimation-error variance, but the small degradation in performance may be worth the computational savings, depending on the application. The bottom figure shows typical error magnitudes for the estimate of the first state for the full-order filter and the reduced-order filter. The reduced-order filter has slightly larger estimation errors.

can be partitioned as follows:

$$P = \begin{bmatrix} \tilde{P} & \Sigma \\ \Sigma^T & \tilde{\tilde{P}} \end{bmatrix} \tag{10.57}$$

We are omitting the time subscripts for ease of notation. The Kalman gain is usually written as $K = P^- H^T (HP^- H^T + R)^{-1}$. With our new notation it can be written as follows:

$$\begin{aligned} K &= \begin{bmatrix} \tilde{K} \\ \tilde{\tilde{K}} \end{bmatrix} \\ &= \begin{bmatrix} \tilde{P}^- & \Sigma^- \\ (\Sigma^-)^T & \tilde{\tilde{P}}^- \end{bmatrix} \times \\ & \begin{bmatrix} H_1^T \\ H_2^T \end{bmatrix} \left[\begin{pmatrix} H_1 & H_2 \end{pmatrix} \begin{pmatrix} \tilde{P}^- & \Sigma^- \\ (\Sigma^-)^T & \tilde{\tilde{P}}^- \end{pmatrix} \begin{pmatrix} H_1^T \\ H_2^T \end{pmatrix} + R \right]^{-1} \end{aligned} \tag{10.58}$$

By multiplying out this equation we can write the formula for $\tilde{K}$ as follows.

$$\tilde{K} = (\tilde{P}^- H_1^T + \Sigma^- H_2^T)\alpha^{-1} \tag{10.59}$$

where α is defined as

$$\alpha = H_1 \tilde{P}^- H_1^T + H_1 \Sigma^- H_2^T + H_2 (\Sigma^-)^T H_1^T + H_2 \tilde{\tilde{P}}^- H_2^T + R \tag{10.60}$$

The measurement-update equation for $\hat{x}$ is normally written as $\hat{x}_k^+ = \hat{x}_k^- + K(y_k - H\hat{x}_k^-)$. With our new notation it is written as

$$\begin{bmatrix} \hat{\tilde{x}}_k^+ \\ \hat{\tilde{\tilde{x}}}_k^+ \end{bmatrix} = \begin{bmatrix} \tilde{K} \\ \tilde{\tilde{K}} \end{bmatrix} \left(y_k - H_1 \hat{\tilde{x}}_k^- - H_2 \hat{\tilde{\tilde{x}}}_k^- \right) \tag{10.61}$$

Since we are not going to estimate $\tilde{\tilde{x}}$ with the reduced-order filter, we set $\hat{\tilde{\tilde{x}}}_k^- = 0$ in the above equation to obtain the following measurement-update equation for $\hat{\tilde{x}}_k^+$:

$$\hat{\tilde{x}}_k^+ = \hat{\tilde{x}}_k^- + \tilde{K}\left(y_k - H_1 \hat{\tilde{x}}_k^- \right) \tag{10.62}$$

The measurement-update equation for P is usually written as $P^+ = (I - KH)P^-(I - KH)^T + KRK^T$. With our new notation it is written as

$$\begin{aligned} \begin{bmatrix} \tilde{P}^+ & \Sigma^+ \\ (\Sigma^+)^T & \tilde{\tilde{P}}^+ \end{bmatrix} &= \left[\begin{pmatrix} I & 0 \\ 0 & I \end{pmatrix} - \begin{pmatrix} \tilde{K} \\ \tilde{\tilde{K}} \end{pmatrix} \begin{pmatrix} H_1 & H_2 \end{pmatrix} \right] \begin{bmatrix} \tilde{P}^- & \Sigma^- \\ (\Sigma^-)^T & \tilde{\tilde{P}}^- \end{bmatrix} \times \\ & \left[\begin{pmatrix} I & 0 \\ 0 & I \end{pmatrix} - \begin{pmatrix} \tilde{K} \\ \tilde{\tilde{K}} \end{pmatrix} \begin{pmatrix} H_1 & H_2 \end{pmatrix} \right]^T + \begin{pmatrix} \tilde{K} \\ \tilde{\tilde{K}} \end{pmatrix} R \begin{pmatrix} \tilde{K}^T & \tilde{\tilde{K}}^T \end{pmatrix} \end{aligned} \tag{10.63}$$

At this point, we assume that $\tilde{\tilde{K}} = 0$. This can be justified if the measurement noise associated with the $\tilde{\tilde{x}}$ states is large, or if H_2 is small, or if the elements of $\tilde{\tilde{x}}$ are small. The $\tilde{\tilde{x}}$ elements are then referred to as consider states, nuisance states, or nuisance variables, because they are only partially used in the reduced-order state estimator, and because we are not interested in estimating them. Based on Equation (10.63), the update equation for $\tilde{P}^+$ can then be written as

$$\begin{aligned} \tilde{P}^+ &= (I - \tilde{K}H_1)\tilde{P}^-(I - \tilde{K}H_1)^T - \tilde{K}H_2(\Sigma^-)^T(I - \tilde{K}H_1)^T - \\ & \quad (I - \tilde{K}H_1)\Sigma^- H_2^T \tilde{K}^T + \tilde{K}H_2\tilde{\tilde{P}}^- H_2^T \tilde{K}^T + \tilde{K}R\tilde{K}^T \end{aligned} \tag{10.64}$$

Multiplying out the above equation and then using the definition of α from Equation (10.60) results in

$$\begin{aligned} \tilde{P}^+ &= \tilde{P}^- - \tilde{K}H_1\tilde{P}^- - \tilde{P}^- H_1^T \tilde{K}^T + \tilde{K}\alpha\tilde{K}^T - \tilde{K}H_2(\Sigma^-)^T - \Sigma^- H_2^T \tilde{K}^T \\ &= \tilde{P}^- - \tilde{K}H_1\tilde{P}^- - \tilde{P}^- H_1^T \tilde{K}^T + (\tilde{P}^- H_1^T + \Sigma^- H_2^T)\tilde{K}^T - \tilde{K}H_2(\Sigma^-)^T - \\ & \quad \Sigma^- H_2^T \tilde{K}^T \\ &= (I - \tilde{K}H_1)\tilde{P}^- - \tilde{K}H_2(\Sigma^-)^T \end{aligned} \tag{10.65}$$

This gives the measurement-update equation for $\tilde{P}^+$. We can go through similar manipulations with Equation (10.63) to obtain

$$\begin{aligned} \Sigma^+ &= (I - \tilde{K}H_1)\Sigma^- - \tilde{K}H_2\tilde{\tilde{P}}^- \\ \tilde{\tilde{P}}^+ &= \tilde{\tilde{P}}^- \end{aligned} \tag{10.66}$$

Putting it all together results in the reduced-order Schmidt–Kalman filter. We can summarize the reduced-order filter as follows.